

SUPERLINEARLY MANY UNIT DISTANCES IN CONVEX POSITION

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ABSTRACT. For every integer $d \geq 2$, we construct a set $P_d \subset \mathbb{R}^2$ in strictly convex position with

$$N_d := |P_d| = 2d^d 2^{d^{d+1}}, \quad u(P_d) \geq \frac{d}{4} N_d,$$

where $u(P)$ counts unordered pairs of distinct points at distance one. Consequently the maximum number of unit distances determined by the vertices of a convex n -gon is not $O(n)$, giving a negative answer to Erdős problem 96. The construction combines a coordinate incidence graph with a two-arc seed near antipodal points. Selected squared distances have errors of order ε^4 ; exact corrections by rotations have size $O(\varepsilon^3)$, while the strict radial support margin is of order ε^2 . A radius perturbation separates all subset products of the correction rotations. The construction, convexity, exact pair count, and negation of the linear bound are formalized in Lean 4 with mathlib.

1. INTRODUCTION

For a finite set $P \subset \mathbb{R}^2$, write

$$u(P) = \#\{\{x, y\} \subseteq P : x \neq y, \|x - y\| = 1\}.$$

We say that P is in *strictly convex position* if $x \notin \text{conv}(P \setminus \{x\})$ for every $x \in P$. Thus every point is a vertex of its convex hull. Let

$$U_c(n) = \max\{u(P) : |P| = n, P \text{ is in strictly convex position}\}.$$

The question attributed to Erdős and Moser, recorded as problem 96 in Bloom's catalogue [2], asks whether $U_c(n) = O(n)$. Edelsbrunner and Hajnal [3] obtained the lower bound $2n - 7$. Aggarwal [1] proved the upper bound $n \log_2 n + O(n)$, improving the earlier upper bound of Füredi. We prove the following.

Theorem 1.1. *For every integer $d \geq 2$, there is a finite set $P_d \subset \mathbb{R}^2$ in strictly convex position such that*

$$|P_d| = N_d := 2d^d 2^{d^{d+1}}, \quad dN_d \leq 4u(P_d). \quad (1.1)$$

Corollary 1.2. *For every $C \in \mathbb{R}$ and $n_0 \in \mathbb{N}$, there is a set P in strictly convex position with $|P| \geq n_0$ and $u(P) > C|P|$. In particular, $U_c(n)$ is not $O(n)$.*

Proof. Choose an integer $d \geq \max\{2, n_0\}$ with $d > 4C$ and apply Theorem 1.1. Since $N_d \geq d$, its set has the required size. The inequality $u(P_d) \geq dN_d/4 > CN_d$ gives the conclusion. $\square$

The use of subsets of commuting rotations is adapted from Swanepoel and Valtr [6]. Their work also uses perturbations and mixed derivatives to separate configurations. The additional ingredient here is a finite incidence seed with a strict radial support margin large enough to preserve every vertex after all the rotations. The argument needs an arbitrarily small parameter for each fixed d ; no bound uniform in d is required.

The proof has three parts. Section 2 gives a general amplification lemma. Sections 3 and 4 construct the seed and its exact corrections. Section 5 proves that the radii can be perturbed so that no two subset products coincide. Section 7 describes the checked Lean statement and the correspondence with the proof below.

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2. RADIAL SUPPORT AND AMPLIFICATION

Identify $\mathbb{R}^2$ isometrically with $\mathbb{C}$, and put

$$\mathcal{H}(a, b) = \operatorname{Re}(\bar{a}(b - a)) = \langle a, b - a \rangle. \quad (2.1)$$

If distinct points b_i satisfy $\mathcal{H}(b_i, b_j) < 0$ for every $i \neq j$, then they are in strictly convex position. Indeed, $\langle b_i, x \rangle < \|b_i\|^2$ holds for all other points and hence for their convex hull, whereas equality holds at b_i .

Lemma 2.1. *Suppose $\|a\|, \|b\| \leq 1$, $|u| = |v| = 1$, and $|u - 1|, |v - 1| \leq H$. Then*

$$\mathcal{H}(au, bv) \leq \mathcal{H}(a, b) + 2H. \quad (2.2)$$

If $a \neq b$ and $\|a\| = \|b\|$, then

$$\mathcal{H}(a, b) = -\frac{1}{2}\|b - a\|^2 < 0. \quad (2.3)$$

Proof. Since $\|au\| = \|a\|$,

$$\mathcal{H}(au, bv) - \mathcal{H}(a, b) = \langle au - a, bv \rangle + \langle a, bv - b \rangle.$$

The absolute value is at most $\|au - a\|\|bv\| + \|a\|\|bv - b\| \leq 2H$. The identity $2\mathcal{H}(a, b) = \|b\|^2 - \|a\|^2 - \|b - a\|^2$ proves the second assertion. $\square$

Let $G = (L \sqcup R, E)$ be a finite simple bipartite graph. Denote the endpoints of $e \in E$ by $\ell(e) \in L$ and $p(e) \in R$, and put $v = |L| + |R|$ and $m = |E|$. Simplicity means that $e \mapsto (\ell(e), p(e))$ is injective.

Lemma 2.2 (Amplification). *Assume that nonzero complex numbers b_i , indexed by $L \sqcup R$, and unit complex numbers w_e , indexed by E , satisfy:*

- (1) *the radii $|b_i|$ are pairwise distinct and at most one;*
- (2) *$|b_{\ell(e)} - b_{p(e)}w_e| = 1$ for every $e \in E$;*
- (3) *with $H = \sum_{e \in E} |w_e - 1|$, $\mathcal{H}(b_i, b_j) < -2H$ for all $i \neq j$;*
- (4) *the products $W_S = \prod_{e \in S} w_e$, $S \subseteq E$, are pairwise distinct.*

If $m \geq 1$, the set

$$B = \{b_i W_S : i \in L \sqcup R, S \subseteq E\} \quad (2.4)$$

is in strictly convex position and satisfies

$$|B| = v2^m, \quad u(B) \geq m2^{m-1}. \quad (2.5)$$

Proof. Taking norms in $b_i W_S = b_j W_T$ gives $|b_i| = |b_j|$ and hence $i = j$. Cancellation, since $b_i \neq 0$, then gives $W_S = W_T$ and $S = T$. Thus the indexing in (2.4) is injective, proving its cardinality.

A telescoping product estimate gives

$$|W_S - 1| \leq \sum_{e \in S} |w_e - 1| \leq H, \quad |W_S| = 1. \quad (2.6)$$

For $i \neq j$, Lemma 2.1 and assumption (3) imply $\mathcal{H}(b_i W_S, b_j W_T) < 0$. For $i = j$ and $S \neq T$, the two points are distinct and have the same radius, so (2.3) gives the same strict inequality. The supporting-line criterion following (2.1) proves strict convexity of the entire set.

For every $e \in E$ and every $S \subseteq E \setminus \{e\}$,

$$|b_{\ell(e)} W_S - b_{p(e)} W_{S \cup \{e\}}| = |b_{\ell(e)} - b_{p(e)} w_e| |W_S| = 1.$$

These are $m2^{m-1}$ distinct unordered pairs. To check this last point, each pair has one endpoint with a label in L and one with a label in R , so equality of unordered pairs cannot reverse the two label classes. Injectivity of the vertex indexing identifies both endpoints, simplicity of G identifies e , and the L endpoint identifies S . $\square$

3. A COORDINATE INCIDENCE GRAPH AND DIGIT SEPARATION

Fix $d \geq 2$ and write $[d] = \{0, \dots, d-1\}$. Choose distinct nonnegative integers h_{ij} for $(i, j) \in [d]^2$, and put $z_{ij} = 5^{h_{ij}}$. For example, one can take $h_{ij} = id + j$. A point label $p \in R$ is a function $p : [d] \rightarrow [d]$. Its cell set is

$$A_p = \{(i, p(i)) : i \in [d]\}.$$

A line label $\ell \in L$ is a coordinate $i \in [d]$ together with a function $q : [d] \setminus \{i\} \rightarrow [d]$, and has cell set

$$A_\ell = \{(j, q(j)) : j \in [d] \setminus \{i\}\}.$$

The edge (p, i) joins p to its restriction away from i . The graph is simple and

$$|R| = d^d, \quad |L| = d d^{d-1} = d^d, \quad v = 2d^d, \quad m = d^{d+1}. \quad (3.1)$$

For a label a , define

$$X_a = \sum_{(i,j) \in A_a} z_{ij}, \quad T_a = 1 + \sum_{(i,j) \in A_a} z_{ij}^2. \quad (3.2)$$

If e completes ℓ to p by adding the cell $(i, p(i))$, then

$$X_p - X_\ell = z_{i,p(i)} > 0, \quad T_p - T_\ell = (X_p - X_\ell)^2. \quad (3.3)$$

Lemma 3.1. *Within either label class, the numbers X_a are pairwise distinct, as are the numbers T_a . For distinct labels a, b in the same class,*

$$|T_b - T_a| < \frac{50}{27}(X_b - X_a)^2. \quad (3.4)$$

Proof. Different labels within either class have different cell sets. Let h be the largest exponent in their symmetric difference, and write $M = 5^h$. The dominant digit and the geometric series give

$$|X_b - X_a| \geq M - \sum_{k < h} 5^k > \frac{3}{4}M, \quad |T_b - T_a| \leq M^2 + \sum_{k < h} 25^k < \frac{25}{24}M^2.$$

Since $(25/24)/(3/4)^2 = 50/27$, these inequalities prove (3.4) and show $X_a \neq X_b$. Applying the same dominant-digit argument in base 25 shows $T_a \neq T_b$. $\square$

4. A SEED WITH EXACT SMALL ROTATION CORRECTIONS

Throughout this section d and all its labels are fixed. Every $O(\varepsilon^k)$ assertion is as $\varepsilon \rightarrow 0^+$, with constants allowed to depend on these finite data. Define

$$C(x, t) = x^2 + \frac{t}{2}, \quad D(x, t) = -x^2 + \frac{t}{2},$$

and, in complex coordinates,

$$a_\ell(\varepsilon) = \varepsilon X_\ell + i\left(-\frac{1}{2} + \varepsilon^2 C(X_\ell, T_\ell)\right), \quad (4.1)$$

$$a_p(\varepsilon) = -\varepsilon X_p + i\left(\frac{1}{2} + \varepsilon^2 D(X_p, T_p)\right). \quad (4.2)$$

These points tend to the two antipodal points $-i/2$ and $i/2$.

Lemma 4.1. *There exist $c > 0$ and $\varepsilon_0 > 0$ such that, for every $0 < \varepsilon < \varepsilon_0$, the seed points are nonzero, have pairwise distinct radii less than one, and satisfy*

$$\mathcal{H}(a_i(\varepsilon), a_j(\varepsilon)) < -c\varepsilon^2 \quad (i \neq j). \quad (4.3)$$

For an incidence edge with $x = X_\ell$, $t = T_\ell$, $y = X_p$, $u = T_p$, the selected squared distance is exactly

$$|a_p(\varepsilon) - a_\ell(\varepsilon)|^2 = 1 + \varepsilon^4 (D(y, u) - C(x, t))^2. \quad (4.4)$$

Proof. For two line labels with data (x, t) and (y, u) , direct expansion gives

$$\mathcal{H}(a_\ell, a_{\ell'}) = \varepsilon^2 \left(-\frac{(y-x)^2}{2} - \frac{u-t}{4} \right) + \varepsilon^4 C(x, t) (C(y, u) - C(x, t)). \quad (4.5)$$

For two point labels the corresponding identity is

$$\mathcal{H}(a_p, a_{p'}) = \varepsilon^2 \left(-\frac{(y-x)^2}{2} + \frac{u-t}{4} \right) + \varepsilon^4 D(x, t) (D(y, u) - D(x, t)). \quad (4.6)$$

By Lemma 3.1, the quadratic coefficient in either formula is strictly less than $-(y-x)^2/27 < 0$. For a pair from opposite classes, in either order, the support tends to $-1/2$. Finiteness now gives a common $c > 0$ and a common interval on which (4.3) holds.

The exact radius formulas are

$$|a_\ell(\varepsilon)|^2 = \frac{1}{4} - \frac{1}{2}\varepsilon^2 T_\ell + \varepsilon^4 C(X_\ell, T_\ell)^2, \quad (4.7)$$

$$|a_p(\varepsilon)|^2 = \frac{1}{4} + \frac{1}{2}\varepsilon^2 T_p + \varepsilon^4 D(X_p, T_p)^2. \quad (4.8)$$

Within one class, unequal T values give a nonzero leading coefficient in every difference of squared radii. Between classes the leading coefficient has magnitude $(T_\ell + T_p)/2 > 0$. Hence the radii are pairwise distinct for small positive ε . They tend to $1/2$, which also proves the nonzero and upper-bound assertions.

Finally, expanding the selected squared distance and using $u-t = (y-x)^2$ from (3.3) cancels its quadratic term and gives (4.4). $\square$

Lemma 4.2. *For each incidence edge $e = (\ell, p)$ there is a continuous choice of unit complex number $w_e(\varepsilon)$, for sufficiently small positive ε , such that*

$$|a_\ell - a_p w_e| = 1, \quad w_e - 1 = O(\varepsilon^3), \quad \text{Im}(\overline{a_\ell} a_p w_e) < 0. \quad (4.9)$$

Proof. Fix an edge and abbreviate $x = X_\ell, t = T_\ell, y = X_p, u = T_p, C = C(x, t), D = D(y, u)$, and $M = D - C$. Put

$$q = \overline{a_\ell} a_p = q_R + i\varepsilon\tau, \quad \tau = \frac{x-y}{2} + \varepsilon^2(xD + yC).$$

Thus $q(0) = -1/4$ and $\tau(0) = (x-y)/2 < 0$. Define

$$F = \tau^2 - \varepsilon^2 M^2 q_R - \frac{1}{4}\varepsilon^6 M^4, \quad w = \frac{q_R + \frac{1}{2}\varepsilon^4 M^2 - i\varepsilon\sqrt{F}}{q}. \quad (4.10)$$

For sufficiently small $\varepsilon > 0, q \neq 0, F > 0$, and $\sqrt{F} - \tau \neq 0$, since their limiting values are respectively $-1/4, \tau(0)^2$, and $-2\tau(0) > 0$. The identity

$$\left(q_R + \frac{1}{2}\varepsilon^4 M^2 \right)^2 + \varepsilon^2 F = q_R^2 + \varepsilon^2 \tau^2 = |q|^2$$

shows $|w| = 1$. Furthermore, $\text{Re}(qw) = q_R + \varepsilon^4 M^2/2$, and therefore

$$|a_\ell - a_p w|^2 = |a_\ell|^2 + |a_p|^2 - 2\text{Re}(qw) = |a_\ell - a_p|^2 - \varepsilon^4 M^2 = 1.$$

The imaginary part of qw is $-\varepsilon\sqrt{F} < 0$.

It remains to verify the order of the correction, since merely obtaining an exact unit distance would not suffice. Rationalization gives the *exact* identity

$$w - 1 = \frac{\varepsilon^3}{q} \left[\frac{\varepsilon M^2}{2} + i \frac{M^2(q_R + \varepsilon^4 M^2/4)}{\sqrt{F} - \tau} \right]. \quad (4.11)$$

Indeed, use $(\sqrt{F} + \tau)(\sqrt{F} - \tau) = F - \tau^2$ in the imaginary part of the numerator of $w - 1$. The bracket divided by q extends continuously to $\varepsilon = 0$, so (4.11) proves the required $O(\varepsilon^3)$ bound. $\square$

Proposition 4.3. *For the graph in Section 3, there are seed points a_i and unit corrections w_e satisfying assumptions (1)–(3) of Lemma 2.2, with strict radius bounds $0 < |a_i| < 1$, and with $\text{Im}(\overline{a_{\ell(e)}} a_{p(e)} w_e) < 0$ for every edge.*

Proof. By finiteness of E , Lemma 4.2 gives $\sum_e |w_e(\varepsilon) - 1| = O(\varepsilon^3)$. Lemma 4.1 gives a common strict support margin $-c\varepsilon^2$. Since $\varepsilon^3 = o(\varepsilon^2)$, choosing one sufficiently small positive ε makes

$$\mathcal{H}(a_i(\varepsilon), a_j(\varepsilon)) < -2 \sum_e |w_e(\varepsilon) - 1| \quad (i \neq j).$$

All the other asserted properties hold on a common sufficiently small interval by the same finiteness argument. Fix an ε in that interval. $\square$

5. SEPARATING SUBSET PRODUCTS BY VARYING RADII

For positive r, s with $-1 < c(r, s) < 1$, define

$$c(r, s) = \frac{r^2 + s^2 - 1}{2rs}, \quad \gamma(r, s) = \arccos c(r, s) \in (0, \pi). \quad (5.1)$$

These inequalities describe the open region in which a nondegenerate triangle with side lengths $r, s, 1$ exists.

Lemma 5.1. *On the region in (5.1),*

$$\frac{\partial^2 \gamma}{\partial r \partial s} = \frac{1}{r^2 s^2 (1 - c(r, s)^2)^{3/2}} > 0. \quad (5.2)$$

Proof. The first and mixed derivatives of c are

$$c_r = \frac{r^2 - s^2 + 1}{2r^2 s}, \quad c_s = \frac{s^2 - r^2 + 1}{2s^2 r}, \quad c_{rs} = -\frac{r^2 + s^2 + 1}{2r^2 s^2}.$$

Differentiating $\gamma_r = -c_r / \sqrt{1 - c^2}$ gives

$$\gamma_{rs} = -\frac{c_{rs}(1 - c^2) + c c_r c_s}{(1 - c^2)^{3/2}}.$$

Substitution and simplification yield $-c_{rs}(1 - c^2) - c c_r c_s = 1/(r^2 s^2)$, proving the formula and its strict positivity. $\square$

Lemma 5.2. *Let $G = (L \sqcup R, E)$ be a finite simple bipartite graph, and let U be a nonempty open set of radius vectors $(r, s) \in (0, \infty)^L \times (0, \infty)^R$ on which every edge pair $(r_{\ell(e)}, s_{p(e)})$ satisfies (5.1). If $(\lambda_e)_{e \in E}$ is a nonzero real coefficient vector, then*

$$A(r, s) = \sum_{e \in E} \lambda_e \gamma(r_{\ell(e)}, s_{p(e)}) \quad (5.3)$$

is not constant on any nonempty open subset of U .

Proof. Choose e_0 with $\lambda_{e_0} \neq 0$. If A were constant on a nonempty open subset, its mixed derivative with respect to $r_{\ell(e_0)}$ and $s_{p(e_0)}$ would vanish there. Every summand other than e_0 depends on at most one of these two coordinates, because the graph is simple. Hence that derivative is

$$\lambda_{e_0} \gamma_{rs}(r_{\ell(e_0)}, s_{p(e_0)}),$$

which is nonzero by Lemma 5.1, a contradiction. $\square$

Lemma 5.3. *With G and U as in Lemma 5.2, fix real constants θ_e and put*

$$w_e(r, s) = \exp(i(\theta_e - \gamma(r_{\ell(e)}, s_{p(e)}))). \quad (5.4)$$

There is a radius vector in U for which the products $\prod_{e \in S} w_e(r, s)$, $S \subseteq E$, are pairwise distinct.

Proof. For distinct subsets $S, T \subseteq E$, put $\lambda_e = \mathbf{1}_S(e) - \mathbf{1}_T(e)$. Their products coincide precisely when

$$\exp(i\Phi(r, s)) = 1, \quad \Phi(r, s) = \sum_e \lambda_e \theta_e - \sum_e \lambda_e \gamma(r_{\ell(e)}, s_{p(e)}). \quad (5.5)$$

The collision set is relatively closed in U by continuity. It has empty interior: otherwise choose a point in its interior and a smaller connected open ball around that point. Since Φ is continuous and takes values in $2\pi\mathbb{Z}$ on that ball, it is constant there, contrary to Lemma 5.2.

There are only finitely many pairs $S \neq T$. Their closed collision sets can be avoided simultaneously in U . Explicitly, starting with a nonempty open subset, remove the first collision set and choose a nonempty open subset of its complement; repeat for each remaining set. The final open subset is nonempty and avoids every collision. $\square$

Proposition 5.4. *The seed and corrections in Proposition 4.3 can be perturbed to satisfy all four assumptions of Lemma 2.2.*

Proof. Write the seed points as $a_i = \rho_i \alpha_i$, where $\rho_i > 0$ and $|\alpha_i| = 1$. Keep the directions α_i fixed and vary the radii independently. For $e = (\ell, p)$, choose any real θ_e with

$$e^{i\theta_e} = \alpha_\ell / \alpha_p.$$

At the original radii, exact unit distance and unit modulus imply

$$\operatorname{Re}(\overline{a_\ell} a_p w_e) = \frac{\rho_\ell^2 + \rho_p^2 - 1}{2}, \quad |\overline{a_\ell} a_p w_e| = \rho_\ell \rho_p.$$

Its strictly negative imaginary part shows that $-1 < c(\rho_\ell, \rho_p) < 1$ and, with the branch in (5.1),

$$\overline{a_\ell} a_p w_e = \rho_\ell \rho_p e^{-i\gamma(\rho_\ell, \rho_p)}. \quad (5.6)$$

Thus the original w_e agrees with (5.4).

For nearby radii (r, s) define $b_\ell = r_\ell \alpha_\ell$ and $b_p = s_p \alpha_p$, and take $w_e(r, s)$ from (5.4). For every feasible radius vector,

$$|b_\ell - b_p w_e(r, s)|^2 = |r_\ell - s_p e^{-i\gamma(r_\ell, s_p)}|^2 = 1.$$

The radii remain positive, pairwise distinct, and less than one on some open neighborhood of the original radius vector. Edge feasibility also persists there. Finally, the strict support inequalities persist:

$$\mathcal{H}(b_i, b_j) < -2 \sum_e |w_e(r, s) - 1| \quad (i \neq j). \quad (5.7)$$

This follows from continuity, the strict inequalities at the original vector, and finiteness of all the index sets.

Choose a nonempty open neighborhood U on which these properties all hold. Lemma 5.3 supplies a vector in U for which all subset products are distinct. At that vector assumptions (1)–(4) of Lemma 2.2 hold simultaneously. $\square$

6. COMPLETION OF THE CONSTRUCTION

Proof of Theorem 1.1. Fix $d \geq 2$ and construct the graph of Section 3. Propositions 4.3 and 5.4 provide seed points and corrections satisfying Lemma 2.2. Its set B , identified with a subset P_d of $\mathbb{R}^2$, is in strictly convex position. By (3.1) and (2.5),

$$|P_d| = 2d^d 2^{d^{d+1}} = N_d, \quad u(P_d) \geq d^{d+1} 2^{d^{d+1}-1} = \frac{d}{4} N_d.$$

This is (1.1). $\square$

Remark 6.1. Every vertex of the amplified set is retained. The proof of convexity does not select a convex subconfiguration: the strict support inequality holds between every two distinct indexed points. Likewise, the distance condition is exact. The $O(\varepsilon^3)$ estimate controls the rotation size only; the corrected distances are identically one.

7. LEAN FORMALIZATION AND REPRODUCIBILITY

The accompanying source project formalizes Theorem 1.1, Corollary 1.2, and the negation of the precise asymptotic statement in the Formal Conjectures library [4]. It uses Lean 4.33.1 and mathlib [5] release v4.33.1, pinned to commit

0df444a360eaa60ab8c11dca51a86af692955474.

All geometric arguments first take place in $\mathbb{C}$, using its ordinary Euclidean norm. The linear isometry `Complex.orthonormalBasisOneI.repr` transfers the result to `EuclideanSpace` over $\mathbb{R}$ with index type `Fin 2`.

The file `Erdos96/ProblemDefinitions.lean` reproduces the four upstream definitions required by the statement: convex independence of a set, the number of unordered unit-distance pairs, the set of possible counts at a fixed cardinality, and its supremum. Their defining bodies are unchanged. The upstream open theorem is not imported. The relevant upstream revision is

b82b08faa9006484021c12005ab41287fb2ffb69.

In particular, the formal convexity predicate is $\forall x \in P, x \notin \text{conv}(P \setminus \{x\})$, and the distance predicate is equality to one. The use of unordered pairs is verified through `Sym2`; repeated endpoints cannot satisfy the unit-distance predicate.

The principal public declarations are:

- `Erdos96.plane_counterexample`: for each $d \geq 2$, a finite subset of the literal Euclidean plane satisfying (1.1);
- `Erdos96.large_convex_counterexample`: the quantified statement of Corollary 1.2;
- `Erdos96.erdos_96_false`: $\neg(U_c(n) = O(n))$ with the upstream supremum definition and the filter `atTop` on $\mathbb{N}$;
- `Erdos96.erdos_96`: the equivalent yes/no statement $\text{False} \leftrightarrow U_c(n) = O(n)$.

The proof structure follows the manuscript. In a few places the formalization uses an equivalent formulation better suited to the library. A finite embedding of the cells into $\mathbb{N}$ supplies the distinct digit exponents. Continuity on a right neighborhood of zero implements the finitely many small-parameter choices. To prove Lemma 5.2, a rectangular finite difference isolates a single edge; two applications of one-variable monotonicity, using (5.2), make that difference strictly nonzero. For the exponential relation, the proof shrinks to a neighborhood where phase differences have absolute value less than π and uses local injectivity of the exponential. These arguments prove the required existence statements without numerical certificates.

The module groups are listed below. Every listed group belongs to the dependency chain of the final result.

Mathematical step	Lean modules in <code>Erdos96/</code>
Radial support and amplification	<code>RadialGeometry</code> , <code>Amplification</code> , <code>Counting</code>
Graph and separated digits	<code>Incidence</code> , <code>Digits</code> , <code>LabelDigits</code>
Seed identities and existence	<code>SeedAlgebra</code> , <code>CorrectionAlgebra</code> , <code>SeedGerms</code> , <code>SeedLimits</code> , <code>SeedPattern</code> , <code>PreparedSeed</code>
Radius perturbation	<code>Angle</code> , <code>GenericAngles</code> , <code>GenericRotations</code> , <code>PhaseGeometry</code> , <code>Perturbation</code>
Full problem statement	<code>Counterexample</code> , <code>ProblemDefinitions</code> , <code>PlaneBridge</code> , <code>Problem</code>

From the Lean project directory, the source and axiom checks are reproduced by

```
lake exe cache get
lake --wfail build
lake env lean Audit.lean
```

The build succeeds with warnings treated as failures. The axiom audit for each principal theorem returns exactly

```
[propext, Classical.choice, Quot.sound]
```

There are no admitted proof obligations or additional axioms in the proof. All 21 project modules additionally pass declaration replay through Lean’s kernel using `leanchecker`. `Audit.lean` prints the public statements, their definitions, and these axiom dependencies. The source archive contains the pinned dependency manifest and the verification logs. Its mathematical conclusion is an existence theorem for every $d \geq 2$; the generic radius choice is noncomputable, and no assertion of a practical coordinate-enumeration algorithm is needed.

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